Konrad E. Gozon

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EDUCATION

The Ohio State University – B.S. Computer Science and Engineering **Columbus, OH | May 2029**

Recipient: Choose Ohio First Future of Microelectronics Scholarship & Engie Axium Scholarship

EXPERIENCE

Software Engineering Intern – Planck AI (Sentinel) **May 2026 – AUG 2026**

Team of 3 | Python, FastAPI, SQLAlchemy, Alembic, Supabase/PostgreSQL, React, TypeScript, Tailwind CSS

- Collaborated on a 3-person engineering team, presenting technical progress and roadmap decisions directly to the CEO and CTO in weekly 1:1 syncs
- Architected and shipped an end-to-end AI red-teaming pipeline that dynamically generates adversarial attack configs across 157 PromptFoo security plugins and executes them against live agent endpoints via async FastAPI background workers, with pre-flight health checks and automated findings persistence; built a lightweight RAG application as a live test target to validate the pipeline end-to-end
- Diagnosed and fixed an async database session bug and a subprocess resource leak in the background job pipeline, adding concurrency limits and process cleanup to prevent orphaned processes and crashes under load
- Led a data-model migration to a unified client/workspace schema across 15+ Alembic and SQL migrations, eliminating cascading crashes across dashboards
- Built a nightly APScheduler-driven scan scheduler with automated Slack reporting (slash commands, run summaries) and delivered core product surfaces in React/TypeScript, shipped across 37 merged pull requests

TECHNICAL SKILLS

Languages: Java, Python, C, x86-64 Assembly, JavaScript (Next.js/React), MATLAB, Shell

Frontend: Next.js 15, React, Tailwind CSS, Zustand, THREE.js, SVG Animation, Recharts

AI / Real-Time: OpenCV, YOLOv11, Depth Anything V2, WebRTC, Gemini 2.0 API, ElevenLabs TTS, MediaPipe, Whisper STT

Backend / DevOps: Supabase, Clerk, Flask, PostgreSQL, NGINX, Amazon S3, Git, PyTest

Tools: Eclipse, PyCharm, VSCode, GDB Debugger, Vim, SFTP, MATLAB Stereo Calibration, ROS2, Linux

HACKATHONS

Splice - AI Reverse Engineering Platform | HackOHI/O 2026

Team of 4 | AWS EC2, Bedrock, S3, Systems Manager, Docker, Ghidra, OpenCode, Freebuff

Built an AI-powered reverse engineering platform that analyzes Electron and native applications, extracts behavioral specifications, and synthesizes reusable software components; won **Most Original at HackOHI/O 2026**.

- Engineered the spec extraction pipeline on AWS Bedrock with Zod-validated structured output, converting reverse-engineered code evidence into typed feature specs with acceptance tests, static checks tying each requirement to source line spans, and live progress streamed to the UI over SSE.
- Developed a sandboxed test oracle using Node.js worker threads and node:vm to run acceptance tests against original and synthesized modules under strict timeouts and call-count checks, paired with an in-memory TypeScript Compiler API typechecker and deterministic Express and persistence generators.
- Established the npm workspace monorepo, a multi-stage Docker build with Docker Compose profiles, and a GitHub Actions CI pipeline, and triaged and merged 15 automated Dependabot dependency updates to keep the codebase current.

HackAI 2026 – AI Interview Coach | HackAI

Team of 4 | Next.js, React, Gemini 2.0, ElevenLabs, MediaPipe, Zustand, Clerk, Supabase

- Engineered an Adaptive Director Mode with dynamic prompt engineering to pivot question difficulty mid-session; implemented Clerk authentication and API gateway proxy bridging the Next.js frontend with a Python AI backend across Gemini 2.0, ElevenLabs, and STT pipelines; placed **7th** among 100+ teams.
- Built a biometric HUD visualizing real-time gaze stability, voice confidence, and fidget index via custom SVG rings and Recharts; synced stress spikes to transcript timestamps for post-interview review.
- Implemented Sigmoid-steered heuristics (tanh-based scoring) for answer quality grading and managed complex async state with Zustand and React Refs to orchestrate turn-taking between Whisper STT, Gemini 2.0, and ElevenLabs TTS without race conditions.

ACTIVITIES

Underwater Robotics Team – Software Sub-Team **Aug 2025 – Present**

– Enhancing robot depth perception by integrating stereo camera input with AI-based depth estimation using Depth Anything V2
– Implementing OpenCV and YOLOv11 for real-time object detection, localization, and depth mapping within frames
– Conducted checkerboard-based stereo camera calibration in MATLAB and deployed calibration parameters (YAML) to the robot via SFTP